



深圳河套學院

Shenzhen Loop Area Institute

Natural Language Processing and Large Language Models

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Shenzhen Loop Area Institute

Autumn, 2026

Lecture 1

Introduction to NLP and LLMs

The Era of AI



The Era of AI



AI Is Not Everything

Stage 3: Neural Network (2010s-2017)

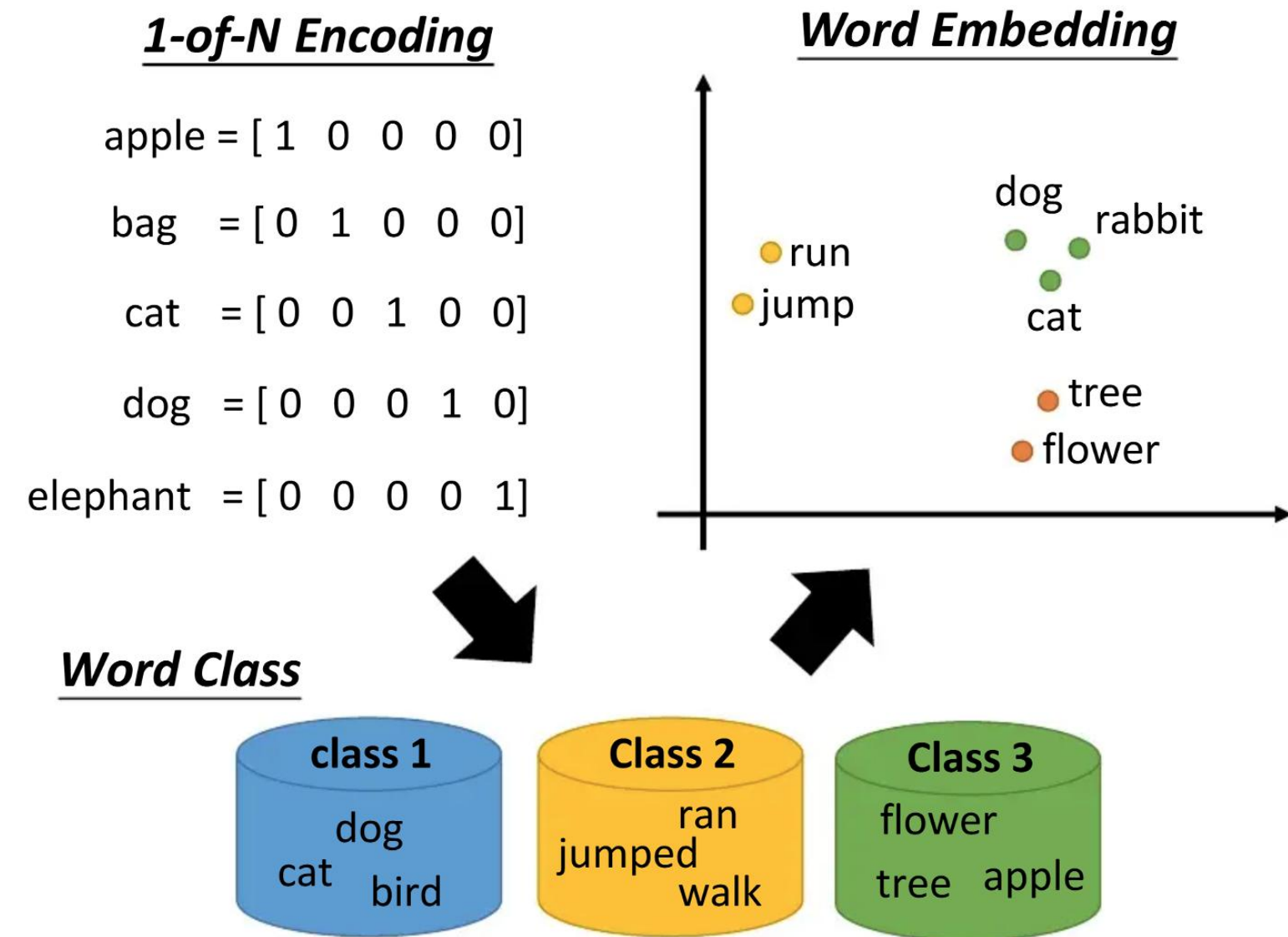
❑ **Word Embeddings:** Maps words into a high-dimensional **vector space**, aiming to capture their **semantic and syntactic meanings**.

■ Words with similar meanings are closer in this space.

❑ Examples: **Word2Vec**, **GloVe**.

Assumption

Words that occur in **similar contexts** tend to share **similar meanings**.

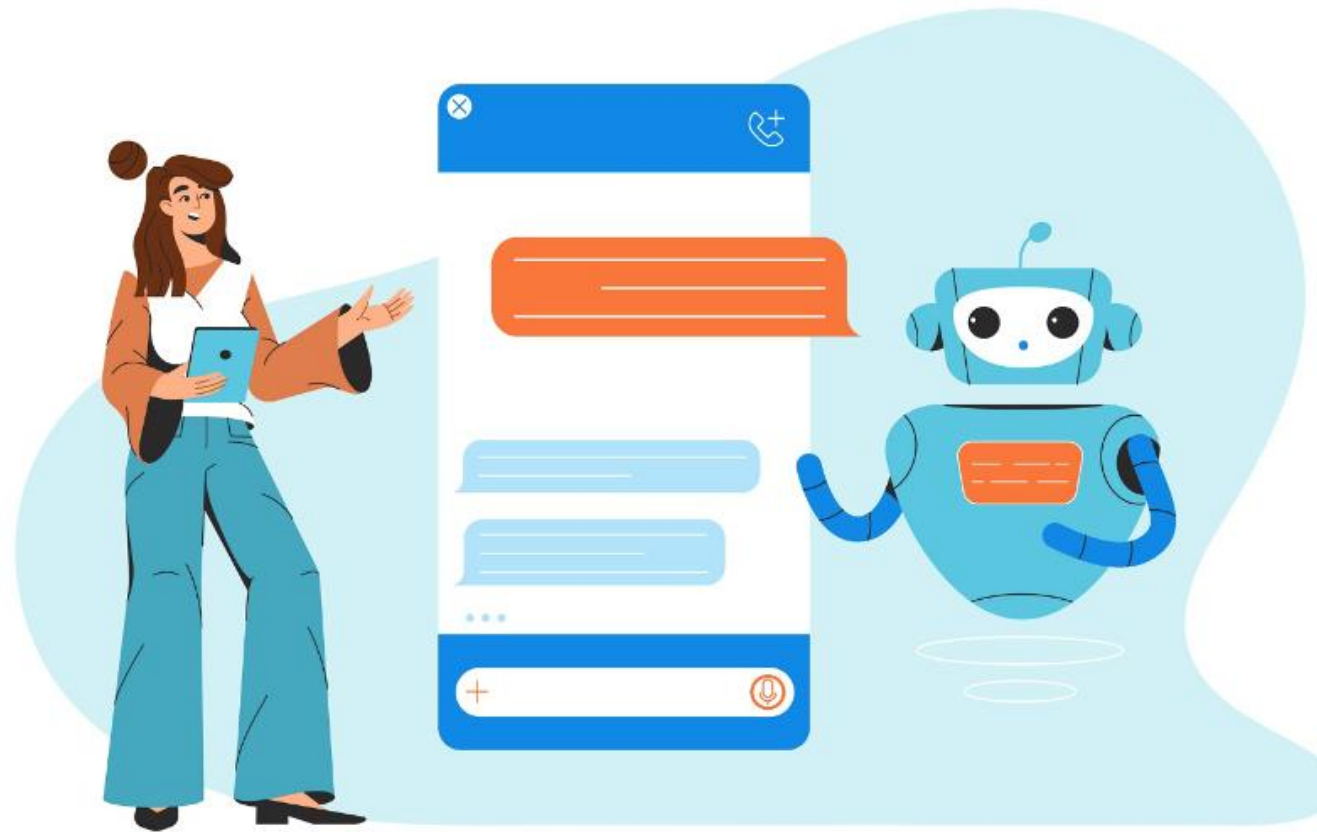


AI Is Not Everything

AI Is Already Part of **Everyday Life**

□ AI is now part of how we **ask, create, code, and communicate.**

■ AI-generated text, software, images, audio, and video are increasingly ordinary.



Takeaway

AI is **reshaping how we live and work.** A new era has arrived.

The result from Codex

AI Is Not Everything

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What is wrong with the AI-generated slide?

Takeaway

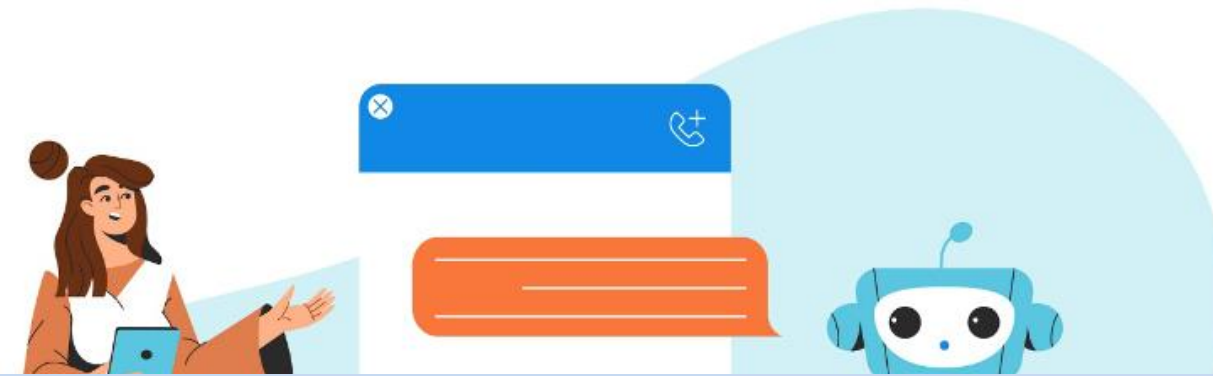
AI is **reshaping how we live and work.** A new era has arrived.

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AI Is Not Everything

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 - AI-generated text, software, images, audio, and video are increasingly ordinary.



AI is useful, but good work still requires judgment?

Takeaway

AI is **reshaping how we live and work.** A new era has arrived.

The result from Codex

Outline

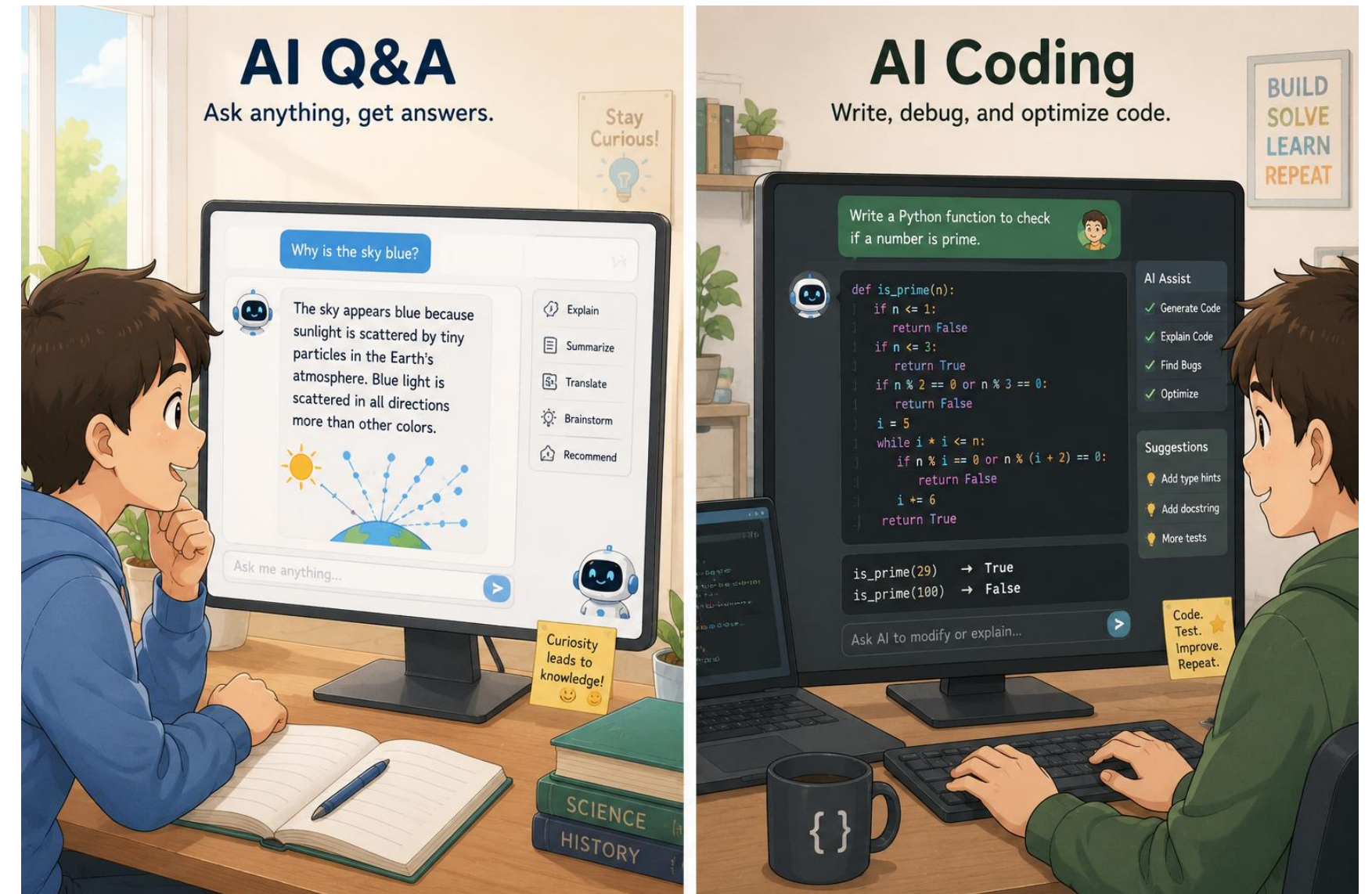
- AI in Everyday Life
- NLP and LLMs
- Course Information
- How LLMs Work
- Practice & Discussion
- Takeaways

Outline

- **AI IN EVERYDAY LIFE**
- NLP and LLMs
- Course Information
- How LLMs Work
- Practice & Discussion
- Takeaways

AI Is Already Part of Everyday Life

- ❑ AI now helps people ask, create, code, and communicate.
- ❑ AI-generated text, images, audio, and video are increasingly ordinary.
- ❑ Examples: ChatGPT, Kimi, Copilot, and text-to-video systems.

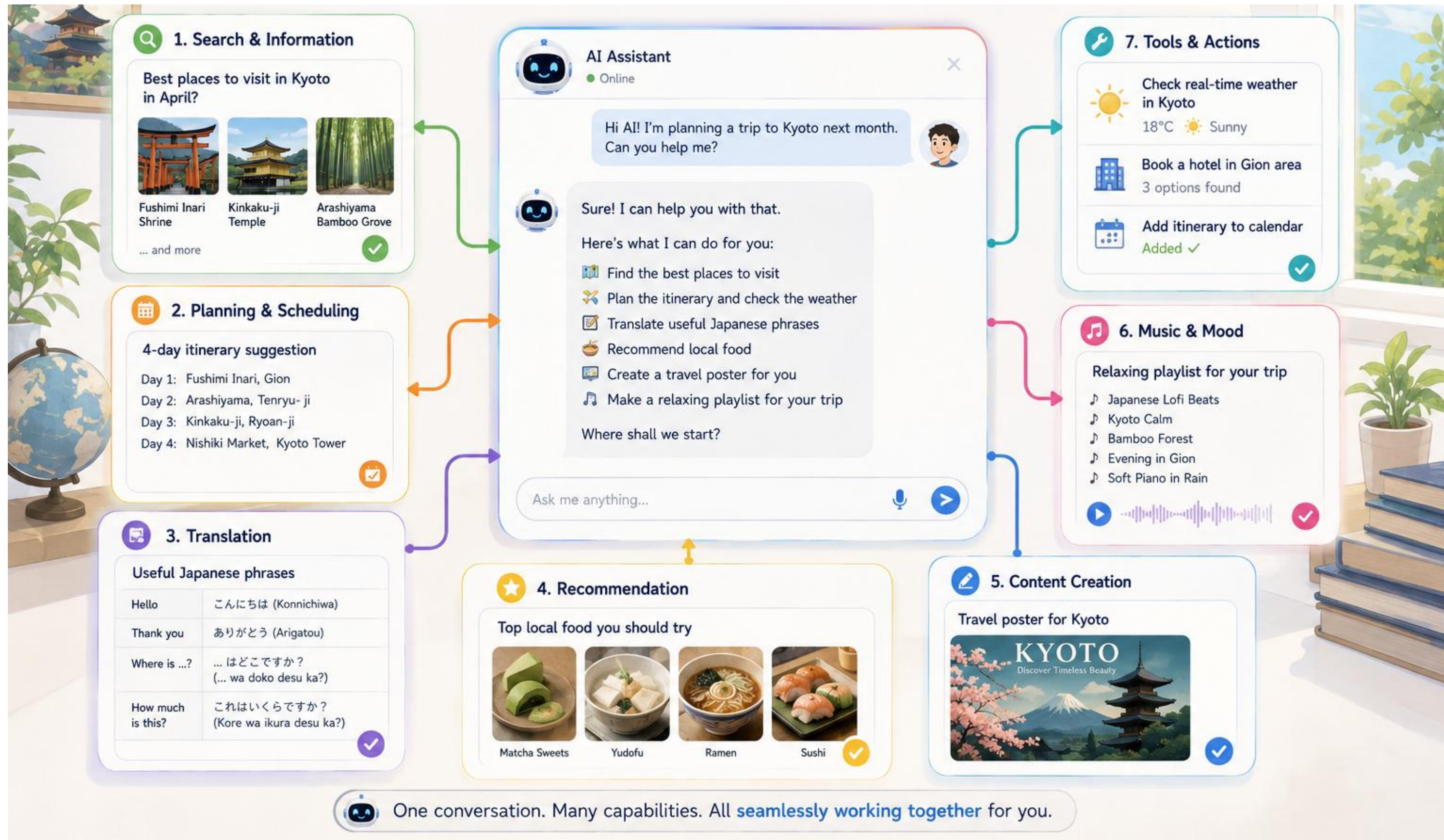


Observation

AI is moving from a specialized tool to an everyday interface.

What Can Today's AI Systems Do?

One conversational AI interface can coordinate several capabilities.



Language Is Becoming a General Interface

- ❑ **Describe a goal in ordinary language.**
 - The system may return text, code, media, or a tool action.
- ❑ Language links the goal, context, tools, and explanation.



Questions?

Why has AI become so powerful only in recent years?

How can one model perform so many different tasks?

Outline

- AI in Everyday Life
- **NLP AND LLMS**
- Course Information
- How LLMs Work
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What Is **Natural Language**?

- ❑ Natural language is how people communicate ideas, emotions, and information.
- ❑ There are around **7,000** languages spoken worldwide.
- ❑ Natural language is flexible, contextual, and shaped by shared knowledge.



Question

Can animal communication be considered natural language?

NO!

Concept: Natural Language

□ Natural Language ≠ Artificial Language

■ Natural Language

- Emerges naturally in human communities
- Flexible, ambiguous, context-dependent

■ Artificial Language

- Designed intentionally for specific purposes
- More explicit and rule-governed

```
! function (t, e) {  
  "object" = typeof exports && "undefined" !== typeof module ? module.exports :  
  "function" = typeof define && define.amd ? define(e) : (t = "undefined"  
  globalThis ? globalThis : t || self).bootstrap = e()  
}(this, (function () {  
  "use strict";  
  const t = {  
    find: (t, e = document.documentElement) => [].concat(...Element.prototype.  
    querySelectorAll.call(e, t)),  
    findOne: (t, e = document.documentElement) => Element.prototype.querySe  
    (e, t),  
    children: (t, e) => [].concat(...t.children).filter(t => t.matches(e)),  
    parents(t, e) {  
      const i = [];  
      let n = t.parentNode;  
      for (; n && n.nodeType === Node.ELEMENT_NODE && 3 < n.nodeType;  
        (e) && i.push(n), n = n.parentNode;
```

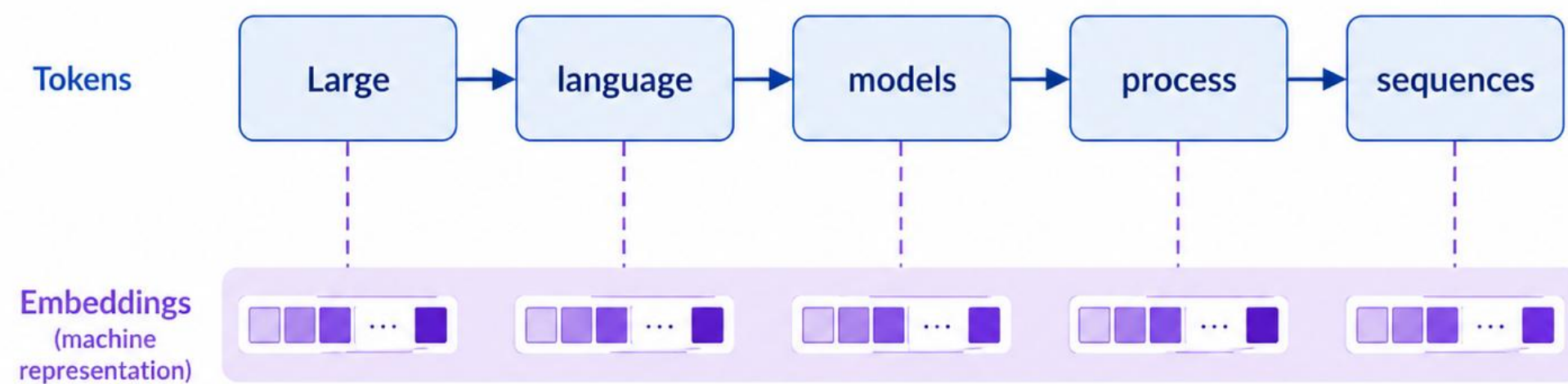


Note

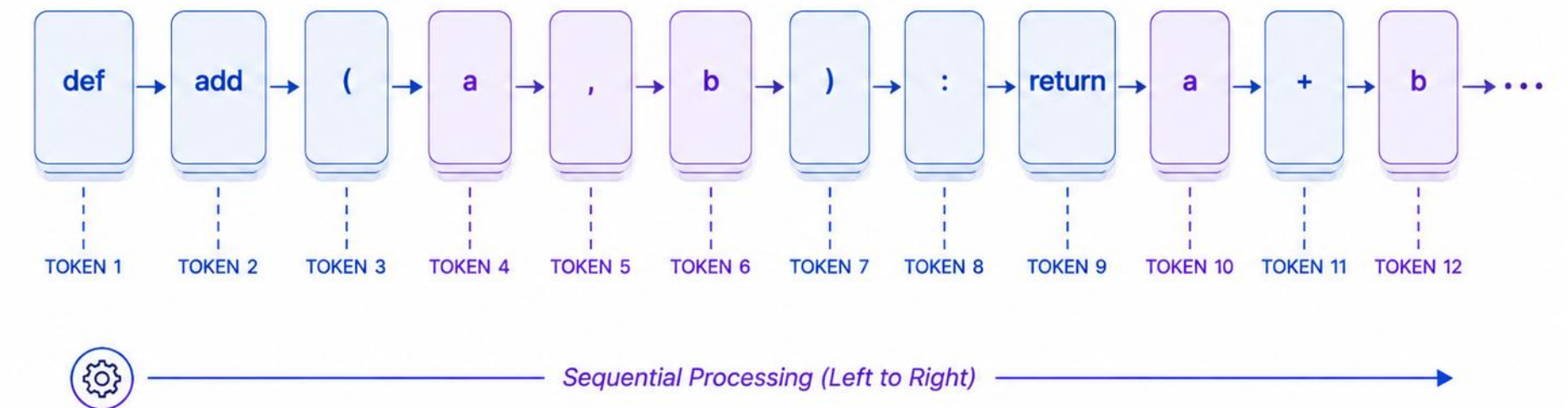
Modern models increasingly operate on sequences beyond text.

Concept: Natural Language

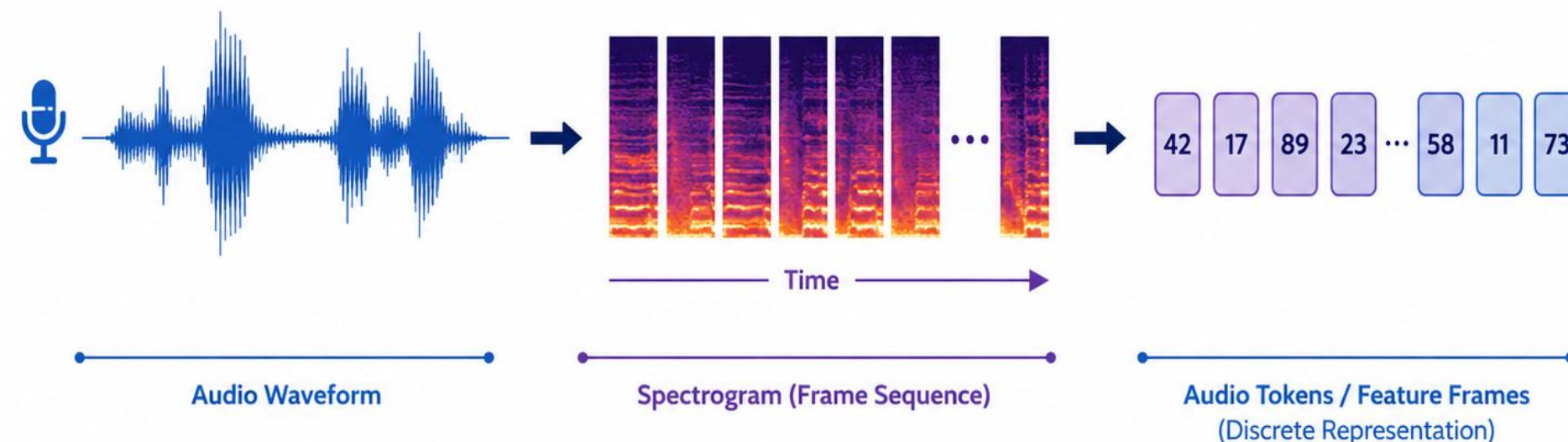
Text Sequence



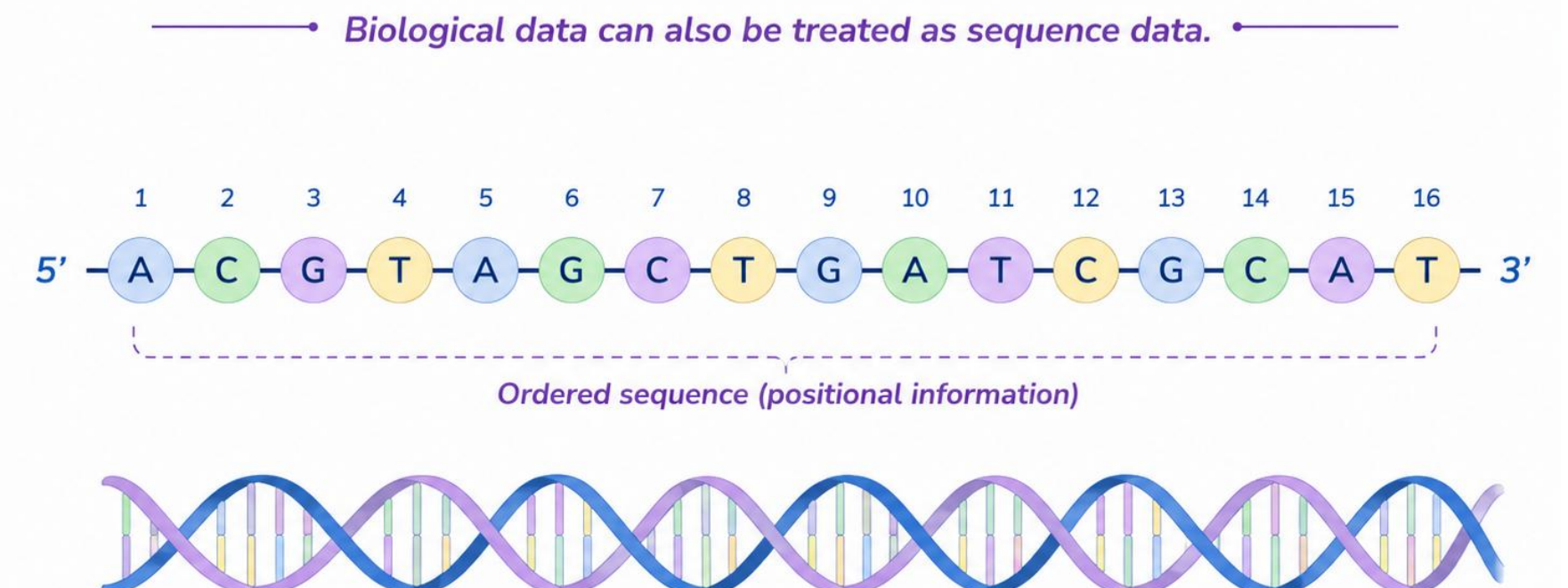
Code Sequence



Speech / Audio Sequence



Biological Sequence



Concept: Natural Language Processing

- NLP = building **computer programs** to analyze, understand and generate **human language - either spoken or written**.
- **Goal**: automate language tasks, augment human capabilities, study language computationally.

Essence (本质)

(*ambiguous*: 模糊的)

Convert *natural language* (complex, ambiguous, human-centric) into a form that computers can *understand, manipulate, and generate*.

(*manipulate*: 操控)

Concept: **Natural Language Processing**

□ **Natural Language Understanding**

- Taking some spoken/typed sentence and working out what it means.
- Examples: *Text Classification, Information Extraction, Sentiment Analysis*.

□ **Natural Language Generation**

- Taking some formal representation of what you want to say and working out a way to express it in a natural (human) language.
- Examples: *Machine Translation, Question Answering, Code Generation*.

Modern LLMs *blur* the traditional boundary between **understanding** and **generation**.

Why Is Language Difficult for Computers?

□ Ambiguity

- One word can have several meanings: *bank, light, match*.
- One sentence can have several structures: "*I saw the student with the telescope.*"

□ **Context-Dependence**: Meaning often relies on the surrounding context.

□ **Flexibility**: The same idea can be expressed in various ways.

□ **Evolvability** (演化): Language changes with the times.

□ **Cultural & Social Context**: Language carries deep cultural connotations.

Ambiguity & Understanding

Meaning depends on context, knowledge, and intention. **Understanding** requires more than matching words to dictionary entries.

Why Is Language Difficult for Computers?

Example 1: Structural Ambiguity

I saw the man with the telescope.



Interpretation A:

I used a telescope to see the man.



Interpretation B:

I saw the man who had a telescope.



Same sentence, different structures, different meanings.

Example 2: Word Sense Ambiguity

He went to the *bank*.



Meaning A:

bank = the side of a river



Meaning B:

bank = a financial institution



A single word can activate very different meanings.

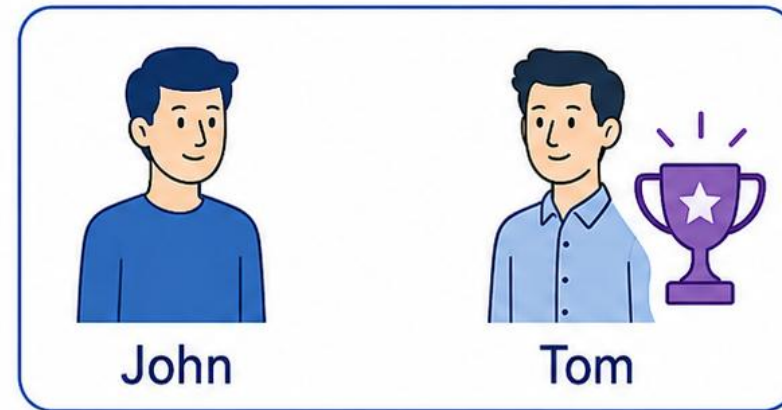
Why Is Language Difficult for Computers?

Example 3: Pronoun / Coreference Ambiguity

John told Tom that *he* won the prize.



Interpretation A: John won.



Interpretation B: Tom won.



Resolving 'he' requires more than grammar alone.

Example 4: Context Changes Meaning

“It is cold here.”



Context A:
a complaint about temperature.



Context B:
an indirect request to close the window / turn off the AC.



Understanding often depends on situation and speaker intent.

Typical NLP Tasks and Applications

□ **Understanding:** *classify, search, extract, answer.*



Understand: classify, search, extract, answer.

1 Classify

Determine the category or intent.

User query:

"I want to cancel my flight."



Intent: CancelFlight

Category: Travel

2 Search

Find relevant information.

Query: "best noise-cancelling headphones under \$200"



Q Top Results:

1. Sony WH-CH720N \$149 ★★★★★
2. Anker Soundcore Q30 \$80 ★★★★★
3. JBL Tune 760NC \$99 ★★★★★

3 Extract

Pull structured information.

Text: "Apple Inc. (AAPL) reported revenue of \$90.8 billion in Q2 2024, up 4% year-over-year."



Company	Apple Inc.
Ticker	AAPL
Metric	Revenue
Value	\$90.8 billion
Period	Q2 2024
YoY Change	+4%

4 Answer

Answer questions accurately.

Q: Why do leaves change color in the fall?



Leaves change color in the fall because trees stop producing chlorophyll as daylight shortens and temperatures drop, revealing other pigments like carotenoids and anthocyanins.



Typical NLP Tasks and Applications

❑ **Generation:** *translate, summarize, transcribe, write, speak, and code.*



What Is a Language Model?

- A **language model** assigns probabilities to token sequences.
 - Given a context, it estimates plausible continuations.
- The output is a distribution over possible continuations.

Model

A **model** is a learned mathematical representation of patterns in language.

$$\hat{y} = f_{\theta}(x)$$

- x : **input** — e.g., a sentence
- $\hat{y}$: **model output / prediction** — e.g., a class label or the next token
- f : **model function / architecture**
- θ : **model parameters learned from data**

What Makes an LLM **Large**?

- "Large" reflects the combined scale of data, parameters, and compute.
 - **Kimi K2**: 1T total parameters, 32B active per token, trained on 15.5T tokens.
 - **DeepSeek-V3**: 671B total parameters, 37B active, trained on 14.8T tokens.



Data

More tokens,
more diverse,
higher quality

+



Parameters

Larger models with
more parameters

+

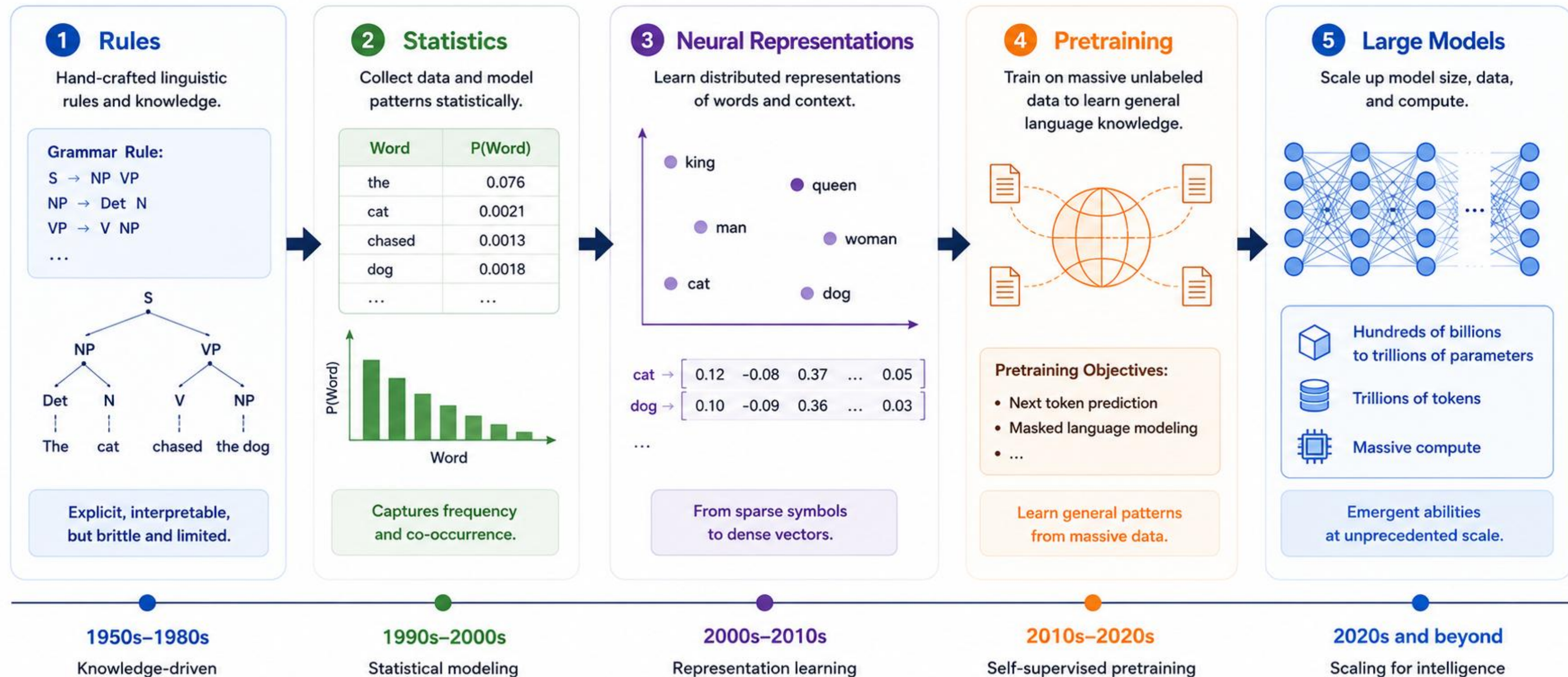


Compute

More FLOPs,
longer training,
more efficient
infrastructure

From Rules to Large Language Models

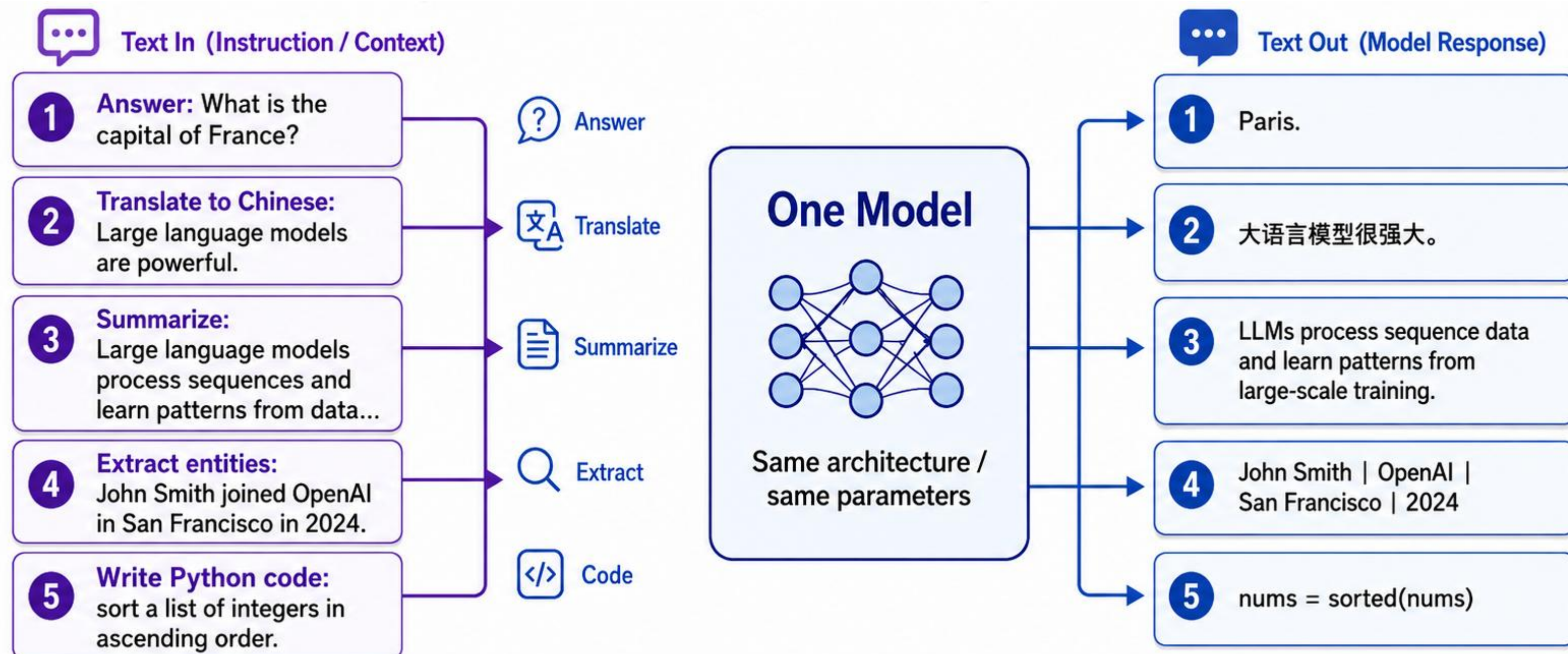
Rules → Statistics → Neural Representations → Pretraining → Large Models



□ **The key shift:** less behavior is written by hand; more is learned from data.

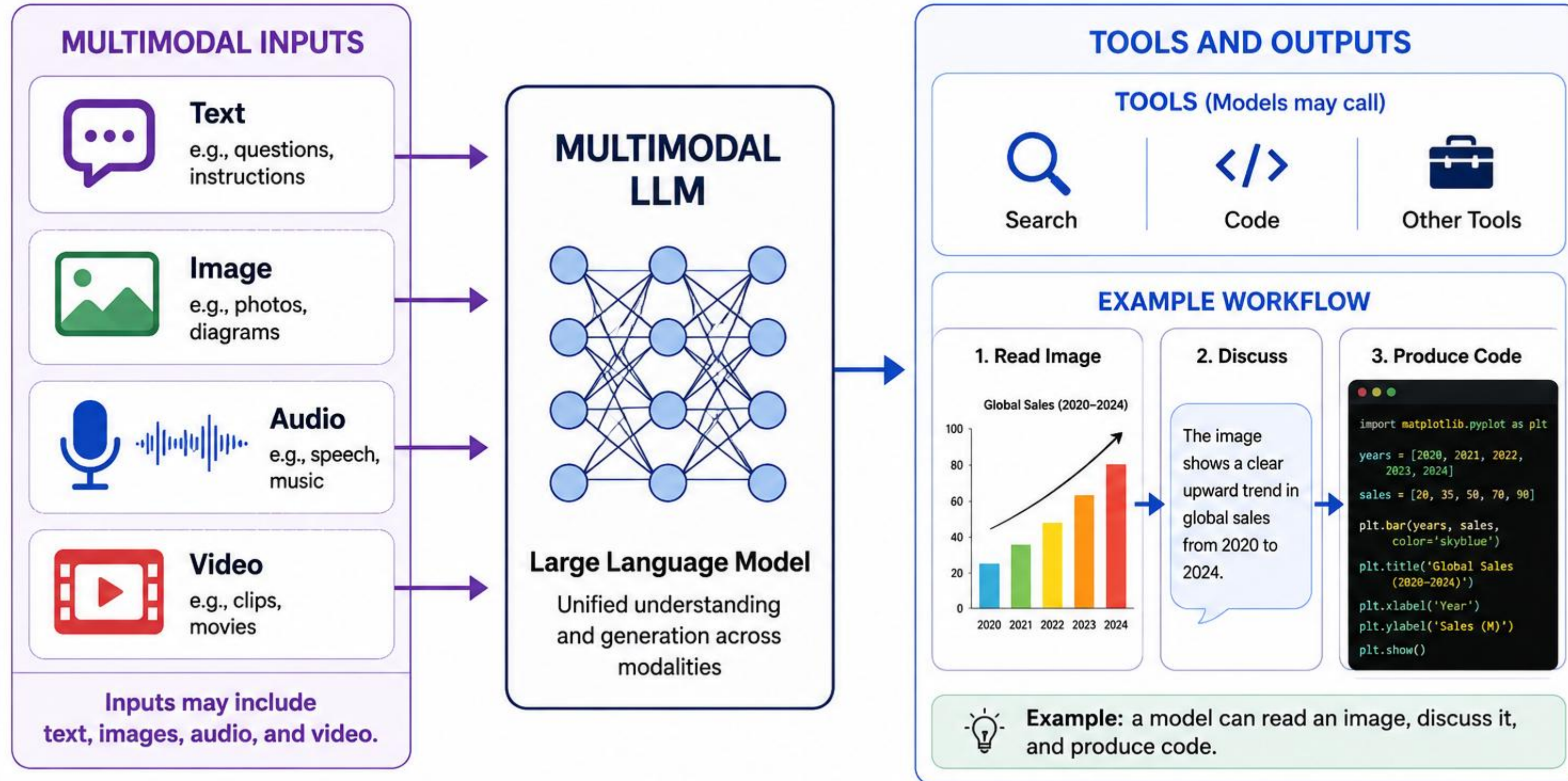
Many Tasks in a **Single Format**

- ❑ The **context** can specify the task: answer, translate, summarize, or code.
- ❑ The model stays the same; the **instruction** specifies the requested behavior.

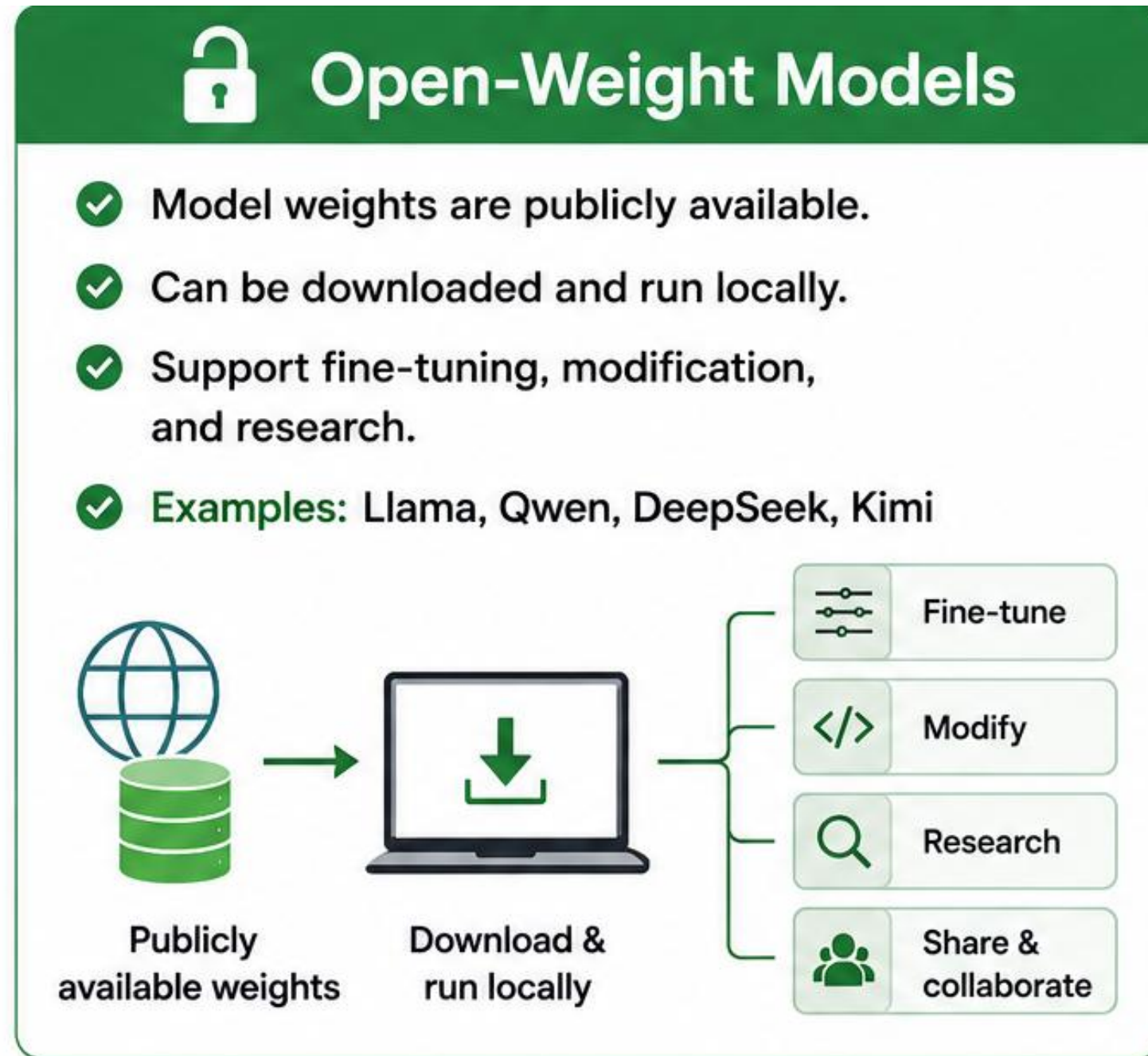


LLMs Are Becoming Multimodal Systems

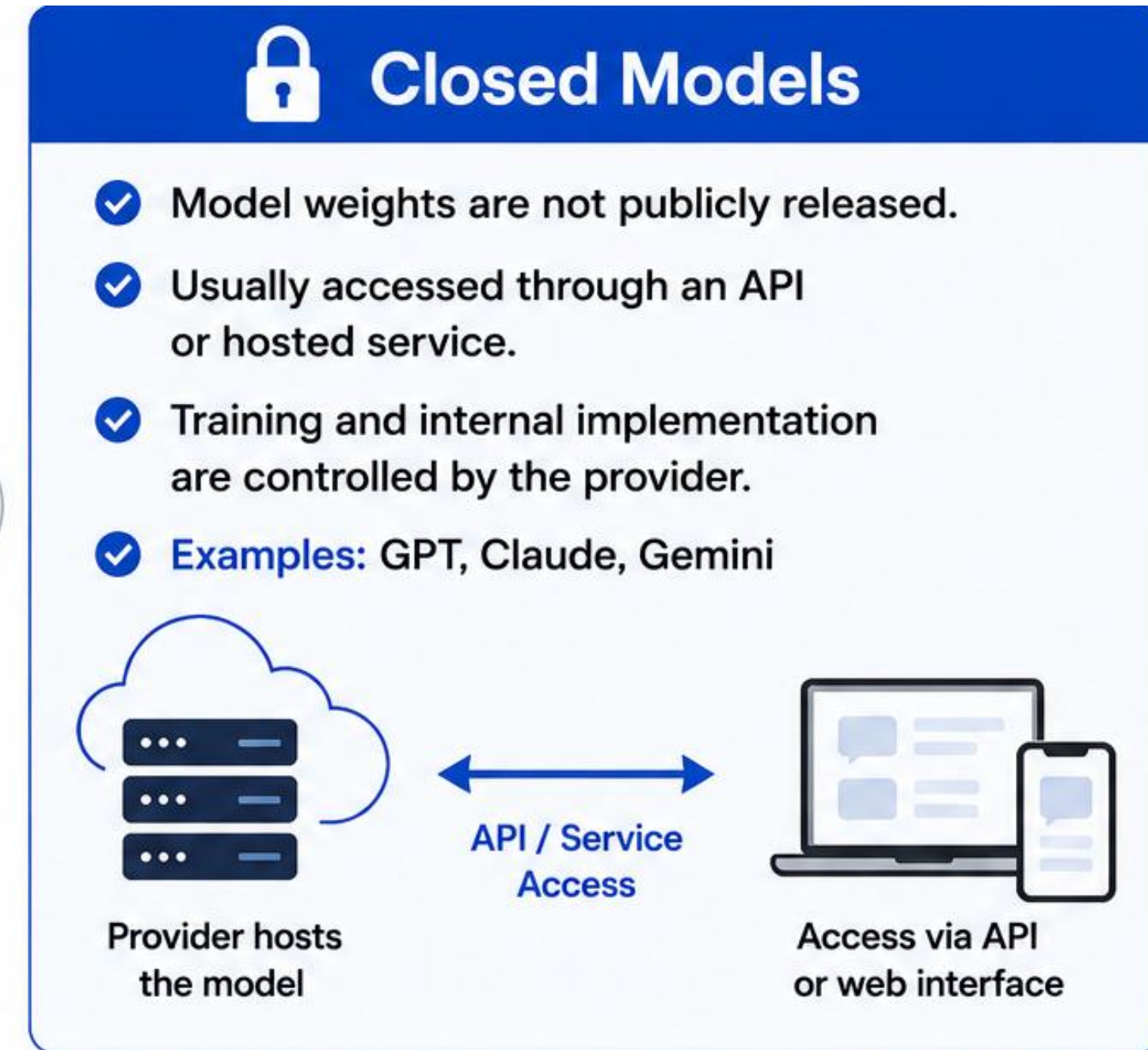
- Inputs may include **text**, **images**, **audio**, and **video**.
- Models may call *search*, *code*, or *other* tools.



Open-weight and Closed Models



VS.



Open-weight $\neq$ open-source.

Open-weight models release the learned weights, but may not release training data, training code, or full training recipes.



Transparency
Higher



Customizability
High



Control
In your hands

VS.



Ease of Use
High



Maintenance
Handled by provider



Privacy
Depends on provider

Outline

- AI in Everyday Life
- NLP and LLMs
- **COURSE INFORMATION**
- How LLMs Work
- Practice & Discussion
- Takeaways

Course Instructors

□ **Xiaojun Quan** – Full-time Professor, SLAI

- PhD, CityUHK.
- Worked at SYSU for 8+ years.
- *NLP, LLMs, Speech LLMs.*
- Email: xiaojunquan@slai.edu.cn



□ **Zhengjun Yue** - Assistant Professor, SLAI

- PhD, Sheffield.
- Worked at Delft University of Technology.
- *Speech and language technology for healthcare.*
- Email: zhengjunyue@slai.edu.cn



We are working on a SLAI focused project on **on-device speech agents**.

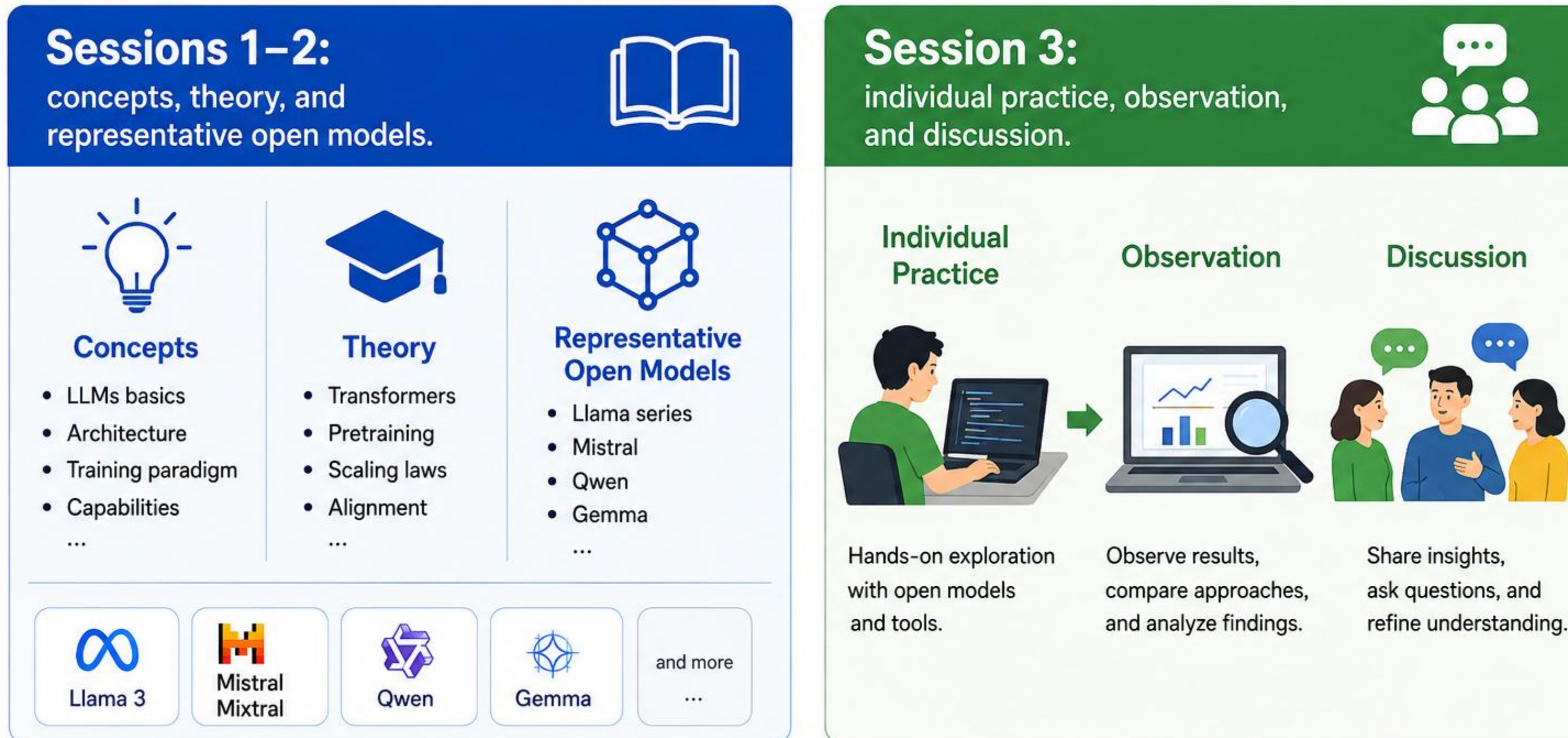
Course Learning Goals

By the end of this course, you should be able to:

1. Understand how modern NLP and LLM systems work, from tokens and Transformers to pretraining, alignment, RAG, agents, and evaluation.
2. Use and adapt representative open-source and API-based language models for real NLP tasks.
3. Build and evaluate practical NLP/LLM applications, and understand why a system works—or fails.
4. Read recent NLP/LLM papers critically and turn research ideas into experiments or course projects.

How Each Class Works: 2 + 1

- ❑ **Sessions 1-2:** concepts, theory, and representative open models.
- ❑ **Session 3:** individual practice, observation, and discussion.



Course Roadmap I

- ❑ **Weeks 1-3:** introduction, tokenization, and embeddings.
- ❑ **Weeks 4-7:** Transformers, BERT/T5/GPT, pretraining, fine-tuning, and alignment.
- ❑ Each block adds concepts needed by the next block.
- ❑ **Instructor:** Xiaojun Quan

Part I

Objective: Build the conceptual and architectural foundation.

Course Roadmap II

- ❑ **Weeks 8-12:** prompting, retrieval, agents, reasoning, and evaluation.
- ❑ **Weeks 13-15:** guest lectures and group project presentations.
- ❑ The final weeks turn those tools into a defensible group project.
- ❑ **Instructor:** Zhengjun Yue

Part II

Study how models interact with information, tools, and evaluation.

Assessment

- ❑ 75% - Final Group Project.
- ❑ 25% - In-class Engagement.
 - In-class engagement includes attendance and participation.

Note

Submitting a project alone is not enough for the highest grade.

What In-class Engagement Means

❑ Come to class and follow the lecture.

- Take part in demonstrations, short tasks, questions, answers, and discussion.
- Engagement is demonstrated during class - not by a separate submission.



References and Tools

□ **Reference sources:**

- *Speech and Language Processing.*
- *Hugging Face documentation, model cards, technical reports, and open-source repositories.*

□ **AI tools:**

- *ChatGPT, Gemini, etc.*

Knowing how to use AI tools well is becoming an essential capability.

WeChat Connection

群聊: 2026NLP&LLMs 课程群

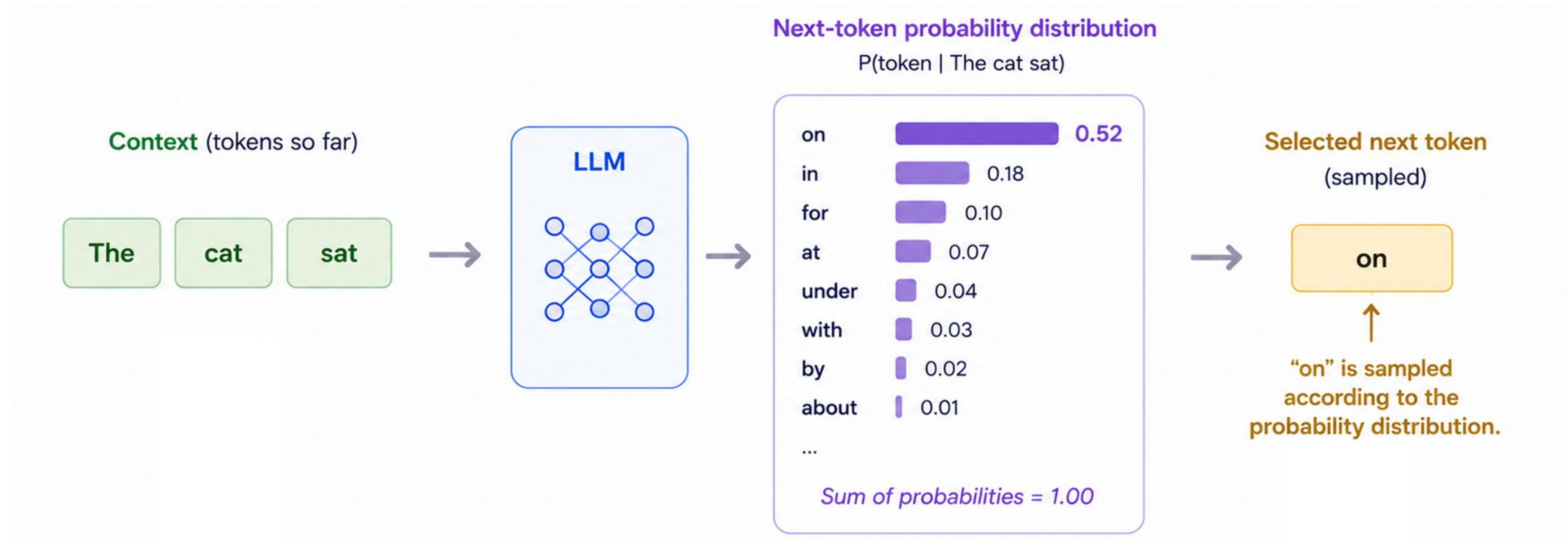


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- NLP and LLMs
- Course Information
- **HOW LLMS WORK**
- Practice & Discussion
- Takeaways

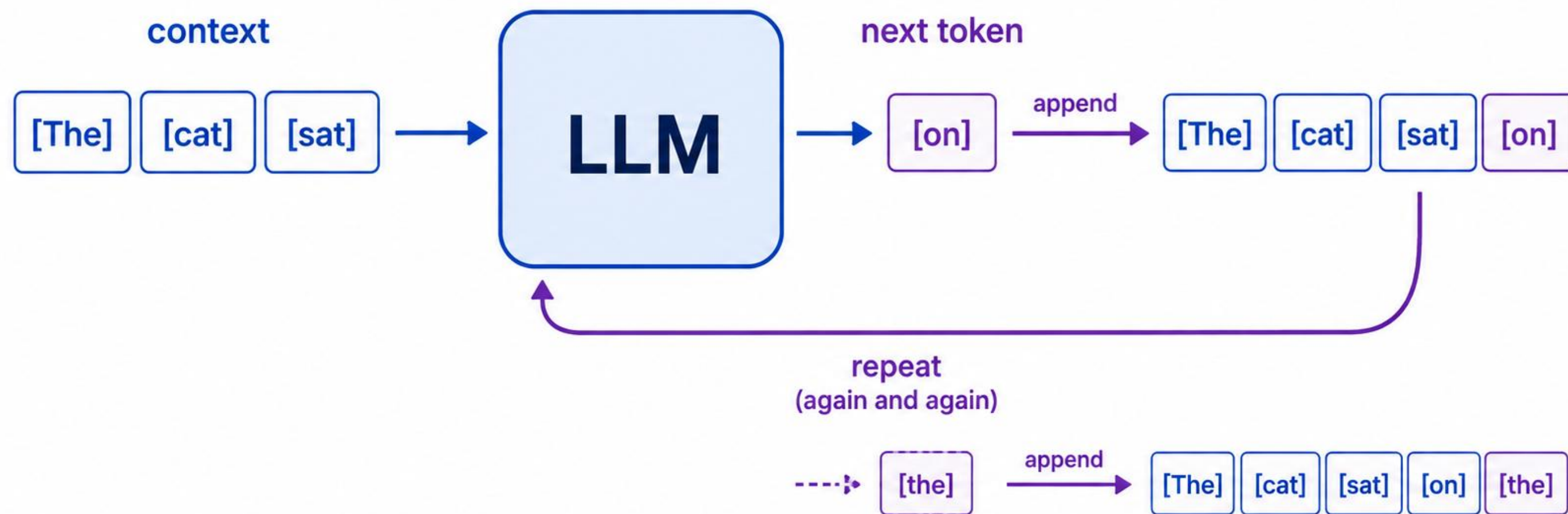
The Core Idea: Predict the Next Token

- Given the tokens/words already in context, the model estimates a probability for the next token/words.



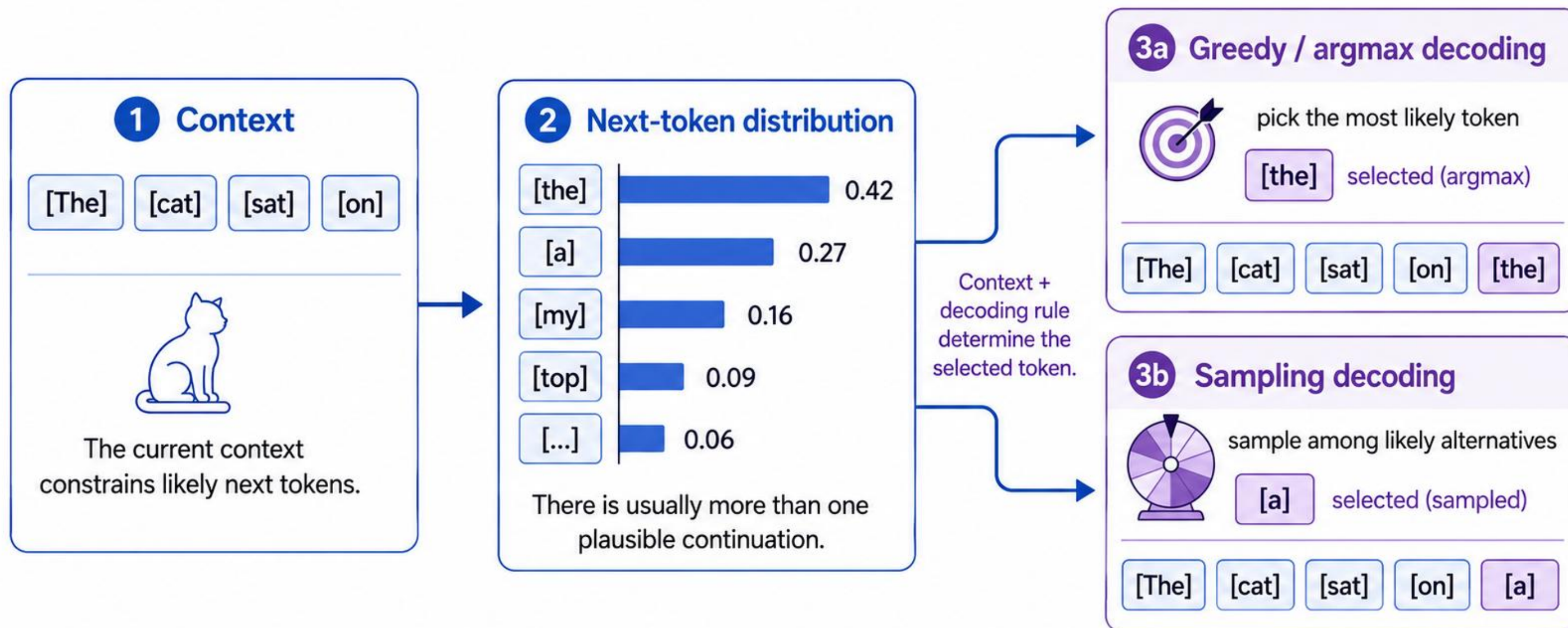
Next-token Generation Is a Loop

- Current context -> probability distribution -> selected token.
 - The selected token becomes part of the next context.
- Generation stops at a stopping token or a configured limit.



Several Next Tokens Are Possible

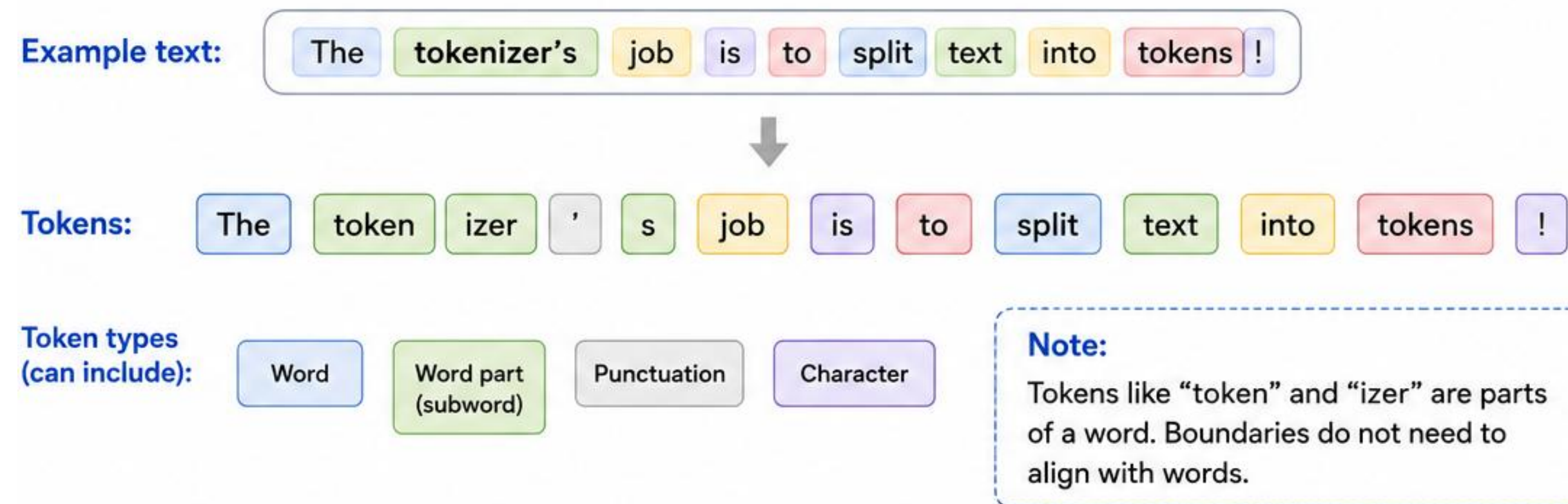
- There is usually **more than one** plausible continuation.
 - Context and the decoding rule shape which token is selected.
- Decoding may choose the **most likely token** or **sample** among alternatives.



The model represents multiple possibilities, not one fixed sentence.

Models Read and Write Tokens

- A **tokenizer** maps text into tokens.
 - A token may be a word, part of a word, punctuation mark, or character.
- **Token boundaries do not necessarily match word boundaries.**

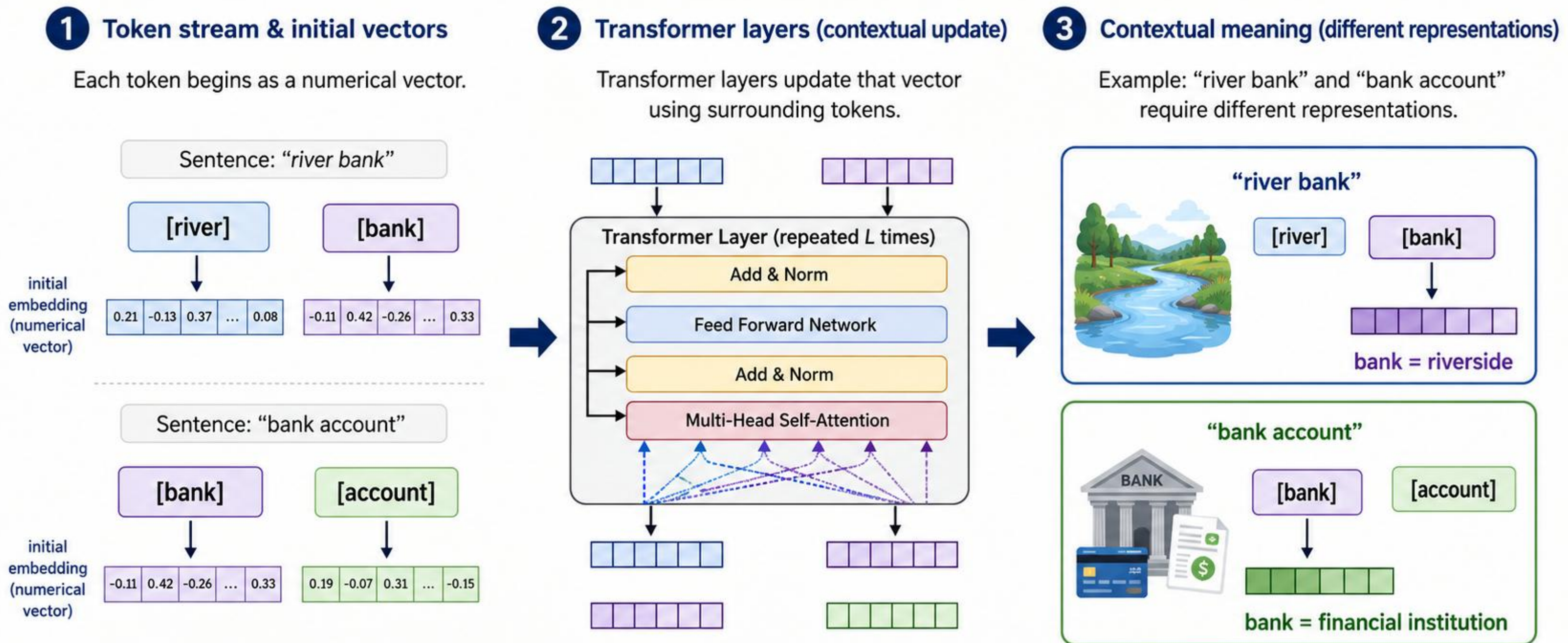


Why tokenization?

Tokenization turns unlimited natural language into a finite vocabulary that neural networks can represent, process, and generate.

Context Creates Representations

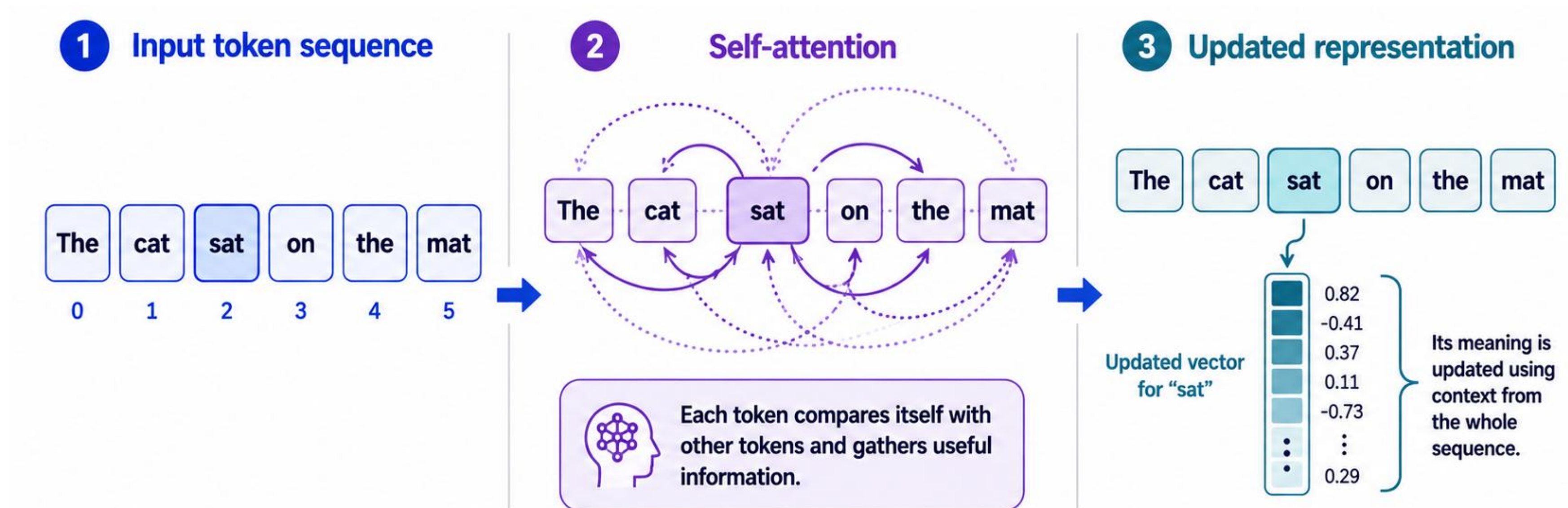
- Each token begins as a numerical vector.
- Transformer layers update that vector using surrounding tokens.



The same token can have different internal representations depending on context.

The Transformer Processes Context

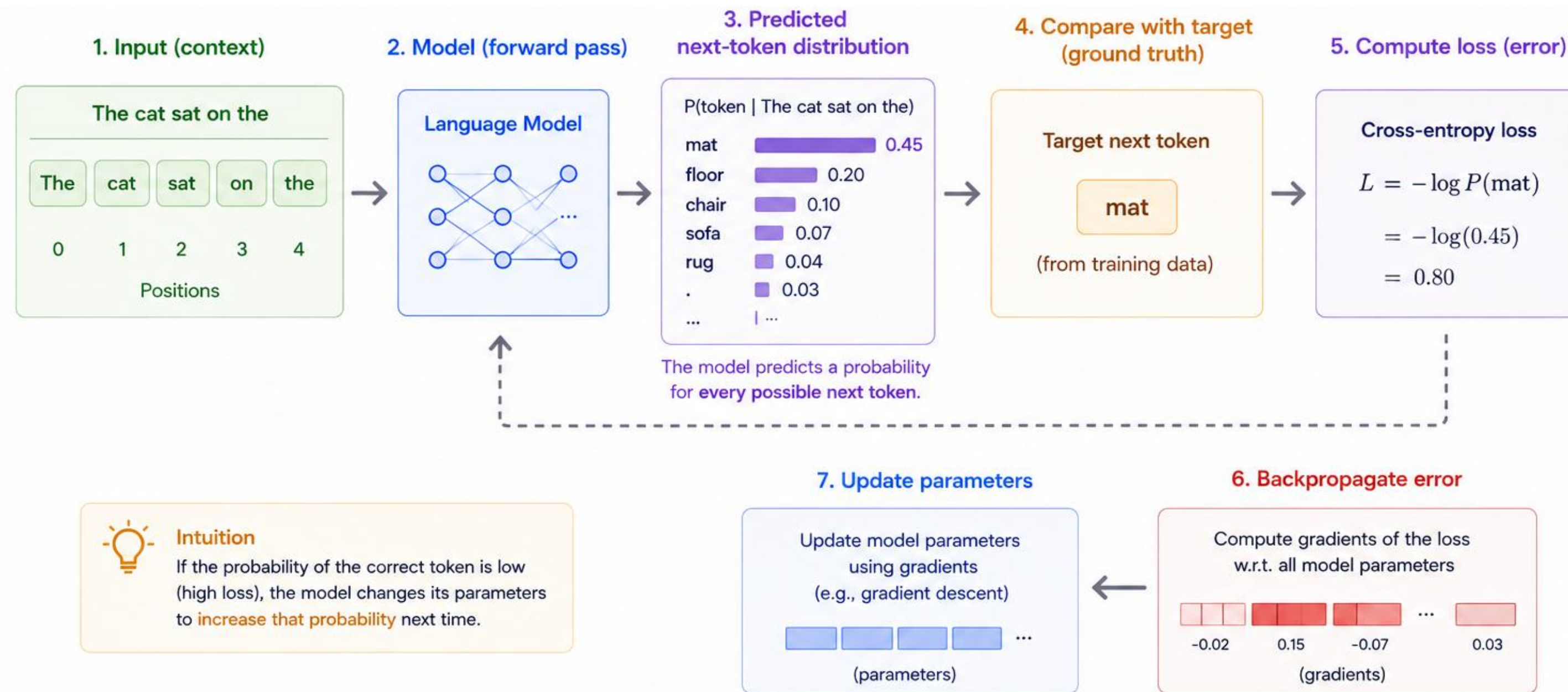
- ❑ **Self-attention** lets each token use information from other positions.



- ❑ Remember: **attention** lets tokens **exchange contextual information**.

Training Reduces Prediction Error

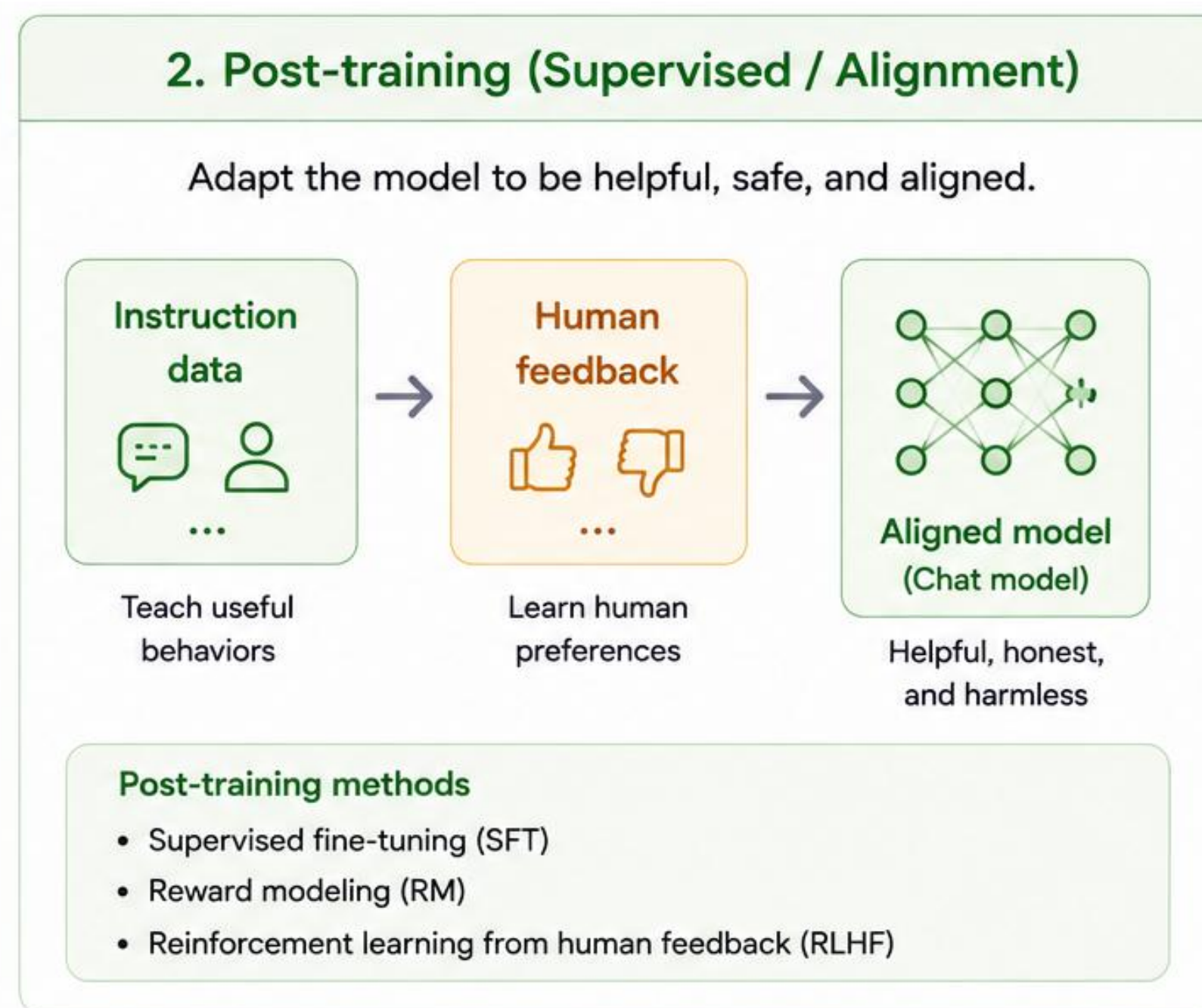
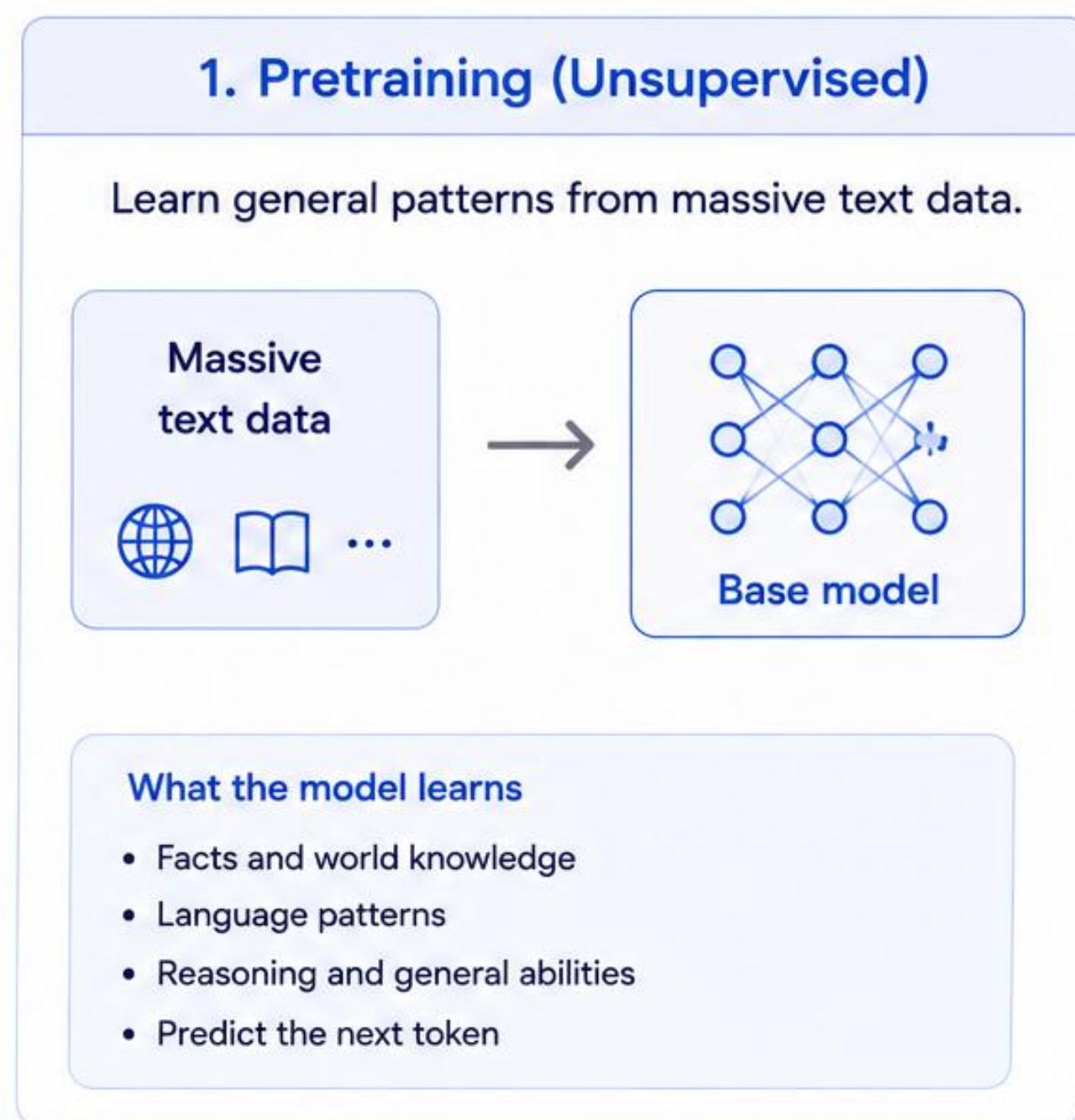
- The model **predicts** a token and compares it with the **training** target.
 - The **error signal** is used to update learned parameters.



- Repeating this process across massive datasets changes the parameters.

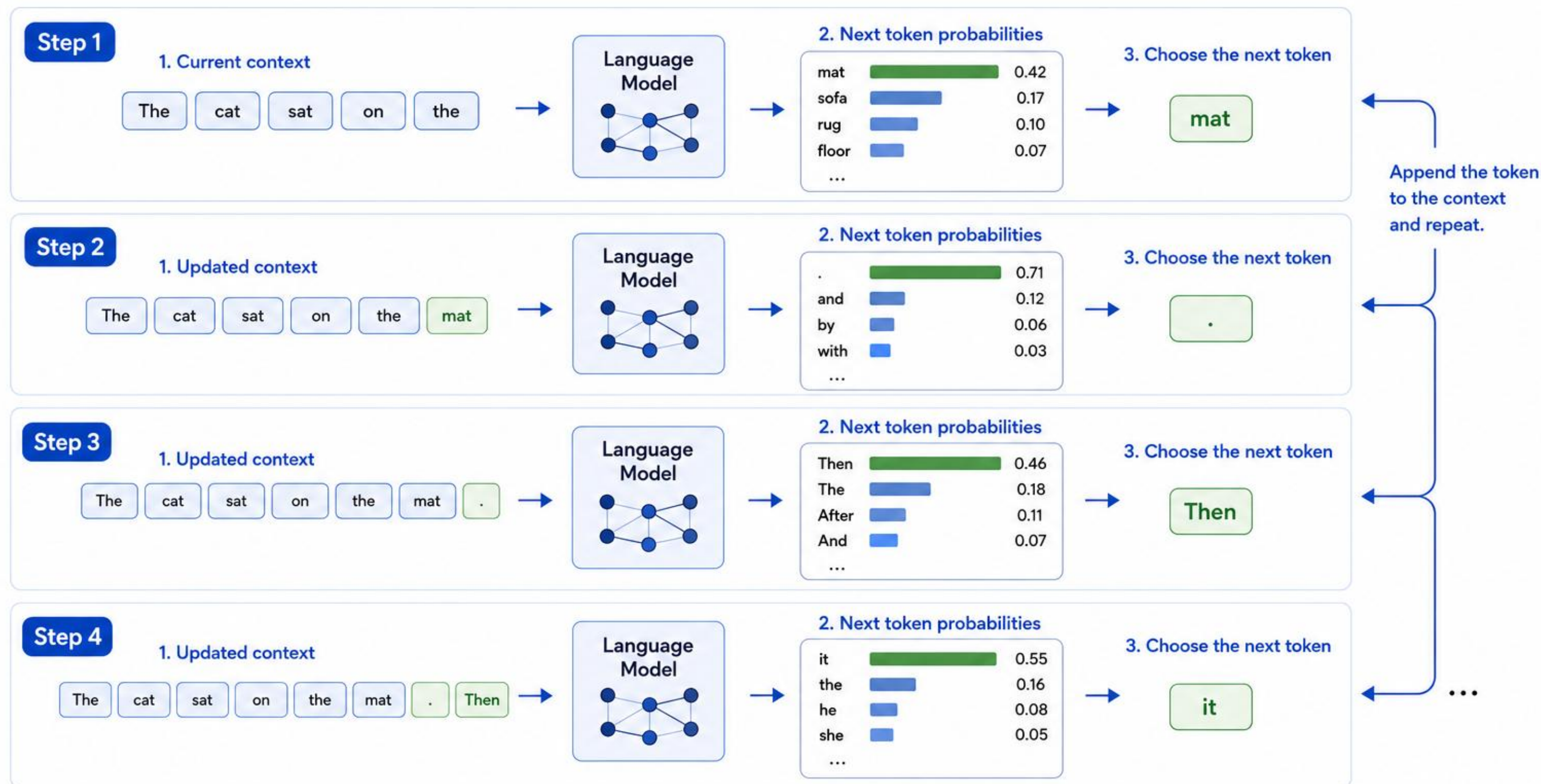
Pretraining and Post-training

- ❑ **Pretraining** learns broad language and code patterns.
- ❑ **Post-training** shapes instruction following and useful assistant behavior.



Inference: One Token at a Time

- During **inference**, the model uses learned parameters.
- The generated token is appended, and the **inference** process repeats.



Training and Inference Are Different

- ❑ **Training** changes parameters using many examples.
- ❑ **Inference** uses fixed parameters to produce an output for the input.
- ❑ The decisive question is **whether the model parameters change**.

Contrast

Training changes the parameters of the model; inference uses them.

Why Prompts and Context Change the Output

- The model's output depends on the context it receives.
 - Changing the context changes the next-token probability distribution.
- The same question can produce different answers under different contexts.

Mechanism

Change the context -> change the probabilities -> change the output.

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- **PRACTICE & DISCUSSION**
- Takeaways

Before Using AI, You Are the Language Model

- **Complete the sentence before opening any AI tool:**
 - "Xiaolin opened the classroom door and found everyone looking at him because ..."

Individual

Write one plausible continuation. Is yours the only reasonable answer?

Same Sentence, Three Different Contexts

- ❑ Context A — Late for class: *The class had already started. The instructor was speaking, and Xiaolin arrived thirty minutes late.*
- ❑ Context B — Birthday surprise: *It was Xiaolin's birthday. His classmates had secretly prepared a small surprise before he arrived.*
- ❑ Context C — New visitor: *Xiaolin had never been to this class before. He was visiting the school for the first time and didn't know anyone in the room.*
- ❑ Keep the unfinished sentence identical across contexts A, B, and C.

How will each context change the likely continuation?

Experiment 1: Change the Context

- Run the three contexts in the same model; keep the unfinished sentence identical.
 - Compare only the first 10-20 words.
- Use the same model and settings so context is the main change.

Experiment 2: Same Prompt, Three Runs

- Use exactly the same prompt three times.
 - Predict first: identical, similar, or different? Then record one pattern and one variation.
- Variation shows that several continuations can remain plausible.

Definition

- **Prompt** = the input that you provide to the model to make it do something.
- **Context** = all information available to the model when predicting the next token.

Discussion: Training or Inference?

- ❑ What changed? Prompt, context, and possibly sampling.
- ❑ What did not change? The model's learned parameters.
- ❑ Changing a prompt changes the input - not the trained parameters.

Answer

Today we observed inference, not training.

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- **TAKEAWAYS**

Takeaways

- ❑ AI is **reshaping** life and work.
- ❑ NLP now extends to **code** and **multimodal** sequences.
- ❑ LLMs learn patterns during **training** and generate token by token during **inference**.
- ❑ **Context** shapes the probability distribution over possible continuations.

 **Learn** something **useful.**
 ***Enjoy*** the process.